

$$e^{j(\omega+2\pi)n} = e^{j\omega n + j2\pi n} = e^{j\omega n} e^{j2\pi n}$$

The DTFT:

- The *discrete-time Fourier transform* (DTFT) is defined as

analysis

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \text{ for } \omega \in \mathbb{R}$$

dot product of $e^{j\omega n}$ and $x[n]$

synthesis

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega,$$

where the notation “ $X(e^{j\omega})$ ” is consistent with the Z-transform. (More on this later...)

The lack of the subscript “ c ” distinguishes the DTFT and Z-transform “ $X(\cdot)$ ” from the CTFT and LT “ $X_c(\cdot)$ ”.

- Essentially, the DTFT gives a “continuous-frequency” characterization of DT signals in the same way that the CTFT gave a “continuous-frequency” characterization of CT signals. (The DTFT/CTFT relationship will be explained further in the context of *sampling*.)
- Important: the DTFT is 2π -periodic in ω :

$$X(e^{j\omega}) = X(e^{j(\omega+2\pi k)}) \text{ for any integer } k,$$

because the function $e^{j\omega}$ is itself 2π -periodic in ω .

In contrast, the CTFT $X_c(j\Omega)$ is not periodic in Ω !

The DTFT — important transform pairs:

- The DTFT is often written in “transform pair” notation

$$x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}).$$

- Some important examples of DTFT transform pairs are:

- Complex exponential:

$$e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi k)$$

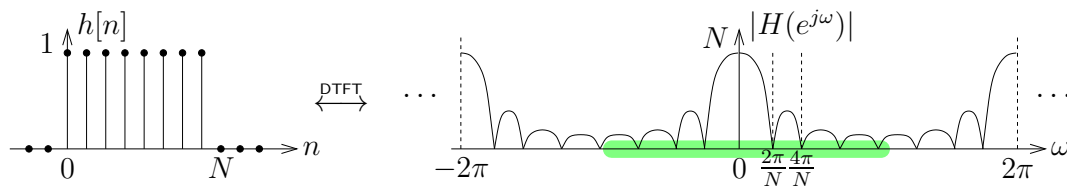
- Kronecker delta:

$$\delta[n - d] \xleftrightarrow{\text{DTFT}} e^{-j\omega d}, d \in \mathbb{Z}$$

or $\omega \in [-\pi, \pi)$

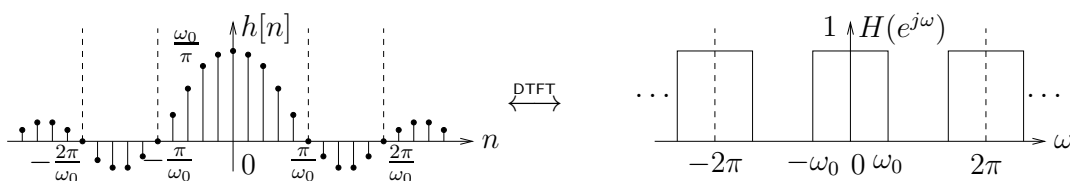
- Rectangle:

$$\begin{cases} 1 & n = 0, \dots, N-1 \\ 0 & \text{else} \end{cases} \xleftrightarrow{\text{DTFT}} e^{-j\omega \frac{N-1}{2}} \frac{\sin(\omega N/2)}{\sin(\omega/2)}$$



- Sinc (with $\omega_0 < \pi$):

$$\frac{\sin(\omega_0 n)}{\pi n} \xleftrightarrow{\text{DTFT}} \begin{cases} 1 & \text{if } |\omega + 2\pi k| < \omega_0, \text{ for } k \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$



The DTFT — important properties:

Below, assume $x[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$ and $w[n] \xleftrightarrow{\text{DTFT}} W(e^{j\omega})$.

1. Linearity:

$$\alpha x[n] + \beta w[n] \xleftrightarrow{\text{DTFT}} \alpha X(e^{j\omega}) + \beta W(e^{j\omega})$$

2. Conjugate symmetry:

$$x^*[n] = x[-n] \Leftrightarrow X(e^{j\omega}) \in \mathbb{R}$$

$$x[n] \in \mathbb{R} \Leftrightarrow X^*(e^{j\omega}) = X(e^{-j\omega})$$

3. Shift/modulation:

$$x[n-d] \xleftrightarrow{\text{DTFT}} e^{-j\omega d} X(e^{j\omega}), d \in \mathbb{Z}$$

$$e^{j\omega_0 n} x[n] \xleftrightarrow{\text{DTFT}} X(e^{j(\omega-\omega_0)})$$

4. Reversal:

$$x[-n] \xleftrightarrow{\text{DTFT}} X(e^{-j\omega})$$

5. Multiplication/convolution:

$$x[n]w[n] \xleftrightarrow{\text{DTFT}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$$

$$\sum_{m=-\infty}^{\infty} x[m]w[n-m] \xleftrightarrow{\text{DTFT}} X(e^{j\omega}) W(e^{j\omega})$$

6. Energy conservation (“Parseval’s theorem”):

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

$$\underbrace{\sum_{n=-\infty}^{\infty} x^*[n] y[n]}_{x[n] \cdot y[n]} = \underbrace{\frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\omega}) Y(e^{j\omega}) d\omega}_{X(e^{j\omega}) \cdot Y(e^{j\omega})}$$

If $x[n] = y[n]$

$$\underbrace{\sum_n |x[n]|^2}_{\text{energy or squared length of } x[n] \text{ as a vector}} = \underbrace{\frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega}_{\text{energy or squared length of } X(e^{j\omega}) \text{ as a vector}}$$

energy or squared length
of $x[n]$ as a vector

energy or squared length
of $X(e^{j\omega})$ as a vector

$$\frac{x[n] \cdot y[n]}{\|x[n]\| \cdot \|y[n]\|} = \frac{X(e^{j\omega}) \cdot Y(e^{j\omega})}{\|X(e^{j\omega})\| \cdot \|Y(e^{j\omega})\|}$$

$$x[n] \longleftrightarrow X(e^{j\omega})$$

ω
English

ω
greek

$$x^*[n] \longleftrightarrow X^*(e^{-j\omega})$$

$$v[n] = x^*[n] y[n] \longleftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

$$z[n] = x^*[n] \quad Z(e^{j\omega}) = X^*(e^{-j\omega})$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{-j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

DC component

$$v[n] = x^*[n] y[n]$$

$$V(e^{j0}) = \sum_{n=-\infty}^{\infty} v[n] e^{-j\omega n} \Big|_{\omega=0} = \sum_{n=-\infty}^{\infty} v[n]$$

||

$$= \sum_{n=-\infty}^{\infty} x^*[n] y[n]$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{-j\theta}) Y(e^{-j\theta}) d\theta \quad \lambda = -\theta$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\lambda}) Y(e^{j\lambda}) d\lambda$$

The DTFT — frequency domain convolution:

- The DTFT-domain convolution *periodic convolution*

$$X(e^{j\omega}) \circledast W(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta$$

differs from “standard” convolution

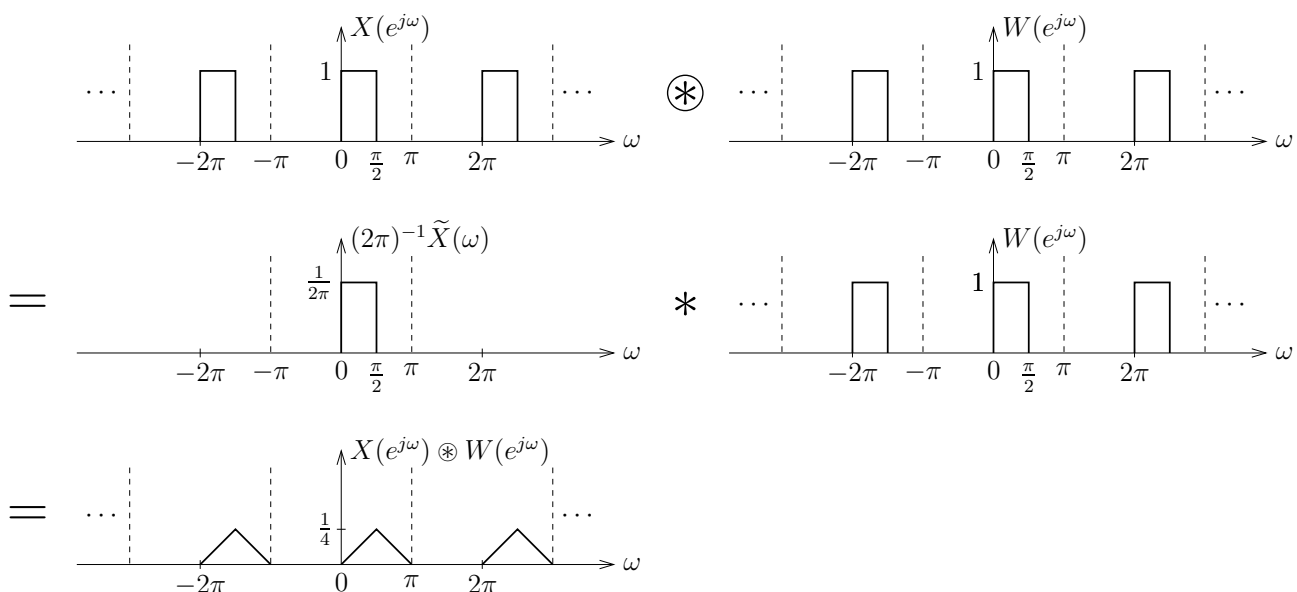
$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$$

in $\frac{1}{2\pi}$ -scaling but also in the limits of integration.

- But, if we define $\tilde{X}(\omega)$ as $X(e^{j\omega})$ with all content outside of $\omega \in (-\pi, \pi]$ suppressed, then we can write

$$\begin{aligned} X(e^{j\omega}) \circledast W(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{X}(\theta) W(e^{j(\omega-\theta)}) d\theta \\ &= \frac{1}{2\pi} \tilde{X}(\omega) * W(e^{j\omega}) \end{aligned}$$

- Example:



Example — DTFT symmetry property:

Example — DTFT convolution property:

$$\begin{aligned}\mathcal{F}_{\text{DTFT}}\left\{\sum_m x[m]w[n-m]\right\} \\&= \sum_n \left(\sum_m x[m]w[n-m]\right)e^{-j\omega n} \\&= \sum_m x[m] \sum_n w[n-m]e^{-j\omega n} \\&= \sum_m x[m] \sum_q w[q]e^{-j\omega(q+m)}\end{aligned}$$

since $q = n - m$ and summation limits don't change

$$\begin{aligned}&= \sum_m x[m]e^{-j\omega m} \sum_q w[q]e^{-j\omega q} \\&= X(e^{j\omega}) W(e^{j\omega})\end{aligned}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \boxed{X(e^{j\omega})} e^{j\omega n} d\omega$$

inverse DTFT
OR
synthesis equation

$$h[n] * x[n] = h[n] * \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) (h[n] * e^{j\omega n}) d\omega$$

$$H(e^{j\omega}) e^{j\omega n}$$

$$\left[h[n] * e^{j\omega n} = \sum_m h[m] e^{j\omega(n-m)} = e^{j\omega n} \underbrace{\sum_m h[m] e^{-j\omega m}}_{H(e^{j\omega})} \right]$$

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \boxed{X(e^{j\omega}) H(e^{j\omega})} e^{j\omega n} d\omega$$

inverse DTFT of $X(e^{j\omega}) H(e^{j\omega})$

Example — DTFT multiplication property:

$$\begin{aligned} & \mathcal{F}_{\text{DTFT}}^{-1} \left\{ \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \right\} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \right) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} W(e^{j(\omega-\xi)}) e^{j\omega n} d\omega \right) d\xi \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) \left(\frac{1}{2\pi} \int_{-\pi-\xi}^{\pi-\xi} W(e^{j\theta}) e^{j(\theta+\xi)n} d\theta \right) d\xi \\ & \qquad \qquad \qquad \text{via } \theta \triangleq \omega - \xi \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) e^{j\xi n} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} W(e^{j\theta}) e^{j\theta n} d\theta \right) d\xi \\ & \qquad \qquad \qquad \text{since DTFT is } 2\pi\text{-periodic} \\ &= x[n]w[n] \\ & \qquad \qquad \qquad \text{via IDTFT definition} \end{aligned}$$

$$\int x(t) e^{-j\omega t} dt \rightarrow \int x(t) e^{-st} dt$$

The Z-transform:

$$s = j\omega + \sigma$$

$$e^s = e^{\sigma + j\omega} = e^{\sigma} e^{j\omega}$$

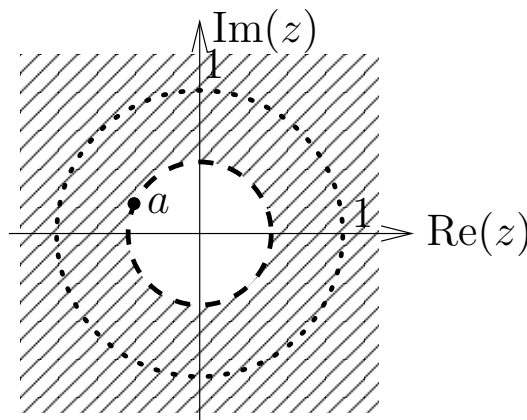
$$z = re^{j\omega}$$

- Recall the 2-sided *Z-transform* (ZT) of $x[n]$:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \text{ for } z \in \mathbb{C} \text{ such that } |X(z)| < \infty.$$

$$X(e^{j\omega}) = \sum_n x[n] e^{-j\omega n} \rightarrow \sum_n x[n] z^{-n}$$

- The DTFT can be considered as a special case of the ZT for which $z = e^{j\omega}$ with $\omega \in \mathbb{R}$, and that is why we write the DTFT of $x[n]$ as “ $X(e^{j\omega})$.”
- The ZT of a given $x[n]$ is defined only for those $z \in \mathbb{C}$ such that $|X(z)| < \infty$. This portion of the complex plane is known as the “*region of convergence* (ROC) of $X(z)$.”
For example, if $x[n] = a^n u[n]$, where $a \in \mathbb{C}$ and $u[n]$ is the step sequence ($u[n] = 1$ for $n \geq 0$ and $u[n] = 0$ otherwise), then one can show that the ROC is $\{z \in \mathbb{C} : |z| > |a|\}$.



$$y(t) = x(t) + a \frac{d}{dt} y(t)$$

$$y[n] = x[n] + a y[n-1]$$

$$\mathcal{L} \quad Y(s) = X(s) + a s Y(s) \quad \Leftrightarrow \quad Y(z) = X(z) + z^{-1} a Y(z)$$

$$\mathcal{FT} \quad Y(\omega) = X(\omega) + a j\omega Y(\omega) \quad \text{bTFT} \quad Y(e^{j\omega}) = X(e^{j\omega}) + e^{-j\omega} a Y(e^{j\omega})$$

$$z = e^s$$

$$Y(s) = X(s) + e^{-s} a Y(s)$$

$$\frac{Y(s)}{X(s)} = \frac{1}{1 - a e^{-s}}$$

$$\frac{Y(s)}{X(s)} = \frac{1}{1 - a e^{-s}}$$

- Like the DTFT, the utility of the ZT stems from the fact that the convolution infinite-sum

$$y[n] = \sum_{m=-\infty}^{\infty} h[m]x[n-m]$$

simplifies to the algebraic expression

$$Y(z) = H(z)X(z)$$

.

- For many signals $x[n]$ (e.g., impulse, step, ramp, geometric signals $a^n u[n]$), there exist convenient closed-form expressions for $X(z)$. Moreover, when $Y(z)$ is rational, its inverse ZT can be calculated using *partial fraction expansion* (PFE) and table-lookup.
- The *advantage* of the ZT vs. the DTFT is that there is a much larger class of signals with elegant transforms. The *disadvantage* of the ZT vs. the DTFT is that the inverse transform step (requiring PFE or contour integration) is more complicated.

In this course, we focus on the DTFT but use the ZT when building block diagrams (since “ z^{-1} ” describes a system that delays its input by one sample).

$$y[n] = ax[n-1] + bx[n-2]$$

Example 1 — LTI transfer functions:

The DTFT leads to a convenient algebra for LTI systems:

- For an LTI system, the impulse response $h[n]$ characterizes the I/O behavior:

$$x[n] \rightarrow \boxed{\mathcal{H}} \rightarrow x[n] * h[n]$$

- The DTFT convolution property says

$$x[n] * h[n] \xleftrightarrow{\text{DTFT}} X(e^{j\omega})H(e^{j\omega}),$$

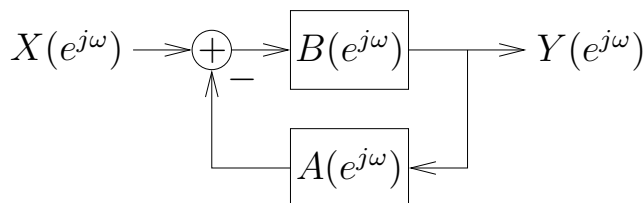
implying that LTI I/O behavior can also be expressed as

$$X(e^{j\omega}) \rightarrow \boxed{\mathcal{H}} \rightarrow X(e^{j\omega})H(e^{j\omega})$$

i.e., *multiplication* by “transfer function” $H(e^{j\omega})$:

$$H(e^{j\omega}) \xleftrightarrow{\text{DTFT}} h[n]$$

- In this way, the DTFT facilitates *algebraic* analysis of networks of interconnected systems!



$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{B(e^{j\omega})}{1 + A(e^{j\omega})B(e^{j\omega})}$$

$$Y = B(X - AY) \quad \rightarrow$$

Example 2 — ARMA transfer function:

Say that we have an “autoregressive moving average” system:

$$y[n] = \sum_{m=0}^M b_m x[n-m] - \sum_{k=1}^N a_k y[n-k].$$

It's not difficult to show that such systems are LTI.

We now derive the *ARMA transfer function*:

$$\begin{aligned} Y(e^{j\omega}) &\triangleq \mathcal{F}_{\text{DTFT}}\{y[n]\} \\ &= \sum_{m=0}^M b_m \mathcal{F}_{\text{DTFT}}\{x[n-m]\} - \sum_{k=1}^N a_k \mathcal{F}_{\text{DTFT}}\{y[n-k]\} \quad (\text{linearity}) \\ &= \sum_{m=0}^M b_m e^{-j\omega m} X(e^{j\omega}) - \sum_{k=1}^N a_k e^{-j\omega k} Y(e^{j\omega}) \quad (\text{delay}) \end{aligned}$$

so that

$$Y(e^{j\omega}) \left(1 + \sum_{k=1}^N a_k e^{-j\omega k} \right) = X(e^{j\omega}) \sum_{m=0}^M b_m e^{-j\omega m}$$

and thus

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{m=0}^M b_m e^{-j\omega m}}{1 + \sum_{k=1}^N a_k e^{-j\omega k}}.$$

Example 3 — Direct form II realization:

Example 4 — LTI response to real sinusoid:

What is the response of an LTI system to a real sinusoid?

$$x[n] = \cos(\omega_0 n + \phi) \rightarrow \boxed{\mathcal{H}} \rightarrow y[n]$$

LTI implies that I/O behavior is characterized by transfer fxn:

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

Using Euler's expansion $\cos(\theta) = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$,

$$X(e^{j\omega}) = \mathcal{F}_{\text{DTFT}} \left\{ \frac{1}{2} (e^{j\omega_0 n + j\phi} + e^{-j\omega_0 n - j\phi}) \right\}$$

Using $e^{\mp j\omega_0 n} \xleftrightarrow{\text{DTFT}} 2\pi \sum_k \delta(\omega \pm \omega_0 + 2\pi k)$,

$$X(e^{j\omega}) = \frac{2\pi}{2} \sum_k \left[e^{j\phi} \delta(\omega - \omega_0 + 2\pi k) + e^{-j\phi} \delta(\omega + \omega_0 + 2\pi k) \right]$$

Thus output is

$$Y(e^{j\omega}) = \pi \sum_k \left(e^{j\phi} \underbrace{H(e^{j\omega})}_{H(e^{j\omega_0})} \delta(\omega - \omega_0 + 2\pi k) + e^{-j\phi} \underbrace{H(e^{j\omega})}_{H(e^{-j\omega_0})} \delta(\omega + \omega_0 + 2\pi k) \right)$$

Using same DTFT pair again,

$$y[n] = \frac{1}{2} \left(e^{j\phi} H(e^{j\omega_0}) e^{j\omega_0 n} + e^{-j\phi} H(e^{-j\omega_0}) e^{-j\omega_0 n} \right)$$

If $h[n] \in \mathbb{R}$, then $H(e^{-j\omega_0}) = H^*(e^{j\omega_0})$. In this case, writing $H(e^{j\omega_0}) = |H(e^{j\omega_0})| e^{j\angle H(e^{j\omega_0})}$, we get

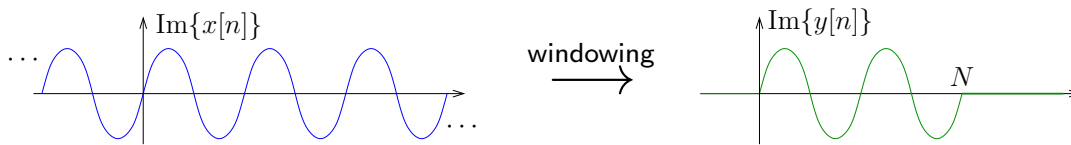
$$\begin{aligned} y[n] &= |H(e^{j\omega_0})| \frac{1}{2} \left(e^{j[\omega_0 n + \phi + \angle H(e^{j\omega_0})]} + e^{-j[\omega_0 n + \phi + \angle H(e^{j\omega_0})]} \right) \\ &= |H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0})) \end{aligned}$$

Real LTI sys changes only gain & phase of real sinusoid!

Example 5 — windowing a sinusoid:

Say $x[n] = e^{j\omega_0 n} \forall n \in \mathbb{Z}$ and $w[n] = \begin{cases} 1 & n = 0, \dots, N-1 \\ 0 & \text{else} \end{cases}$

and $y[n] = w[n]x[n]$:



What is $Y(e^{j\omega})$?

Recall that $X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi k)$.

The DTFT multiplication property then says

$$\begin{aligned}
 Y(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\xi}) W(e^{j(\omega-\xi)}) d\xi \\
 &= \sum_{k=-\infty}^{\infty} \int_{-\pi}^{\pi} \delta(\xi - \omega_0 + 2\pi k) W(e^{j(\omega-\xi)}) d\xi \\
 &= \int_{-\pi}^{\pi} \delta(\xi - \omega_0) W(e^{j(\omega-\xi)}) d\xi \quad \text{assuming } \omega_0 \in (-\pi, \pi] \\
 &= W(e^{j(\omega-\omega_0)}).
 \end{aligned}$$

For the (causal N -length) rectangular “window” $w[n]$, know

$$W(e^{j\omega}) = e^{-j\omega \frac{N-1}{2}} \frac{\sin(\omega N/2)}{\sin(\omega/2)}.$$

